Name: ________________________
Score: ______ / ______

Quiz 3

Part 1: Quiz 3

1

Consider a scenario where the following infinite matrix of bits is given, with all except those on the diagonal are covered by black cards.

$$
\begin{matrix}
0 & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \cdots \\
\blacksquare & 1 & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \cdots \\
\blacksquare & \blacksquare & 0 & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \cdots \\
\blacksquare & \blacksquare & \blacksquare & 1 & \blacksquare & \blacksquare & \blacksquare & \cdots \\
\blacksquare & \blacksquare & \blacksquare & \blacksquare & 0 & \blacksquare & \blacksquare & \cdots \\
\blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare & 1 & \blacksquare & \cdots \\
\blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare & 0 & \cdots \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots
\end{matrix}
$$

Here, each row is a (countably) infinitely long binary string and the number of rows is (countably) infinite. All bits except those on the diagonal are covered by black squares, and the bits on the diagonal alternate between 0 and 1, starting with a 0.
What would an infinitely long binary string look like if it has the property that when all the black cards are removed it is guaranteed to be different from every row?

A.
0101010...

B.
1101010...

C.
1010100...

D.
None of the above

Answer Point Value: 1.0 points
Answer Key: D

2

Is the following a valid encoding of a Turing machine M based on the encoding we discussed in class? (Note that we are not including the encoding of the input string w).

1010110111011001010111011101

True

False

Answer Point Value: 1.0 points
Answer Key: True

3

There are languages that are decidable but not regular, so the set of decidable languages must have a larger cardinality than the set of regular languages.

True

False

Answer Point Value: 1.0 points
Answer Key: False

4

The number of Turing machines we can design is uncountable.

True

False

Answer Point Value: 1.0 points
Answer Key: False

5

For languages A, B and C, if A $\leq_T$ B and B $\leq_T$ C and C is decidable then A is decidable.

True

False

Answer Point Value: 1.0 points
Answer Key: True

6

For languages A, B and C, if A $\leq_T$ B and B $\leq_T$ C and A is decidable then C is decidable.

True

False

Answer Point Value: 1.0 points
Answer Key: False

7

The HALT language is not Decidable because complement of HALT is Turing-Recognizable and it is not co-Turing recognizable.

True

False

Answer Point Value: 1.0 points
Answer Key: False

8

If A $\leq_T$ B and B is a regular language, then A is a regular language.

True

False

Answer Point Value: 1.0 points
Answer Key: False

9

For every computable function \(f\), there exists a constant \(c_f\) such that \(K (f (x)) \leq K (x)+c_f \)
for all \(x\), where \(K\) is Kolmogorov complexity.

True

False

Answer Point Value: 1.0 points
Answer Key: True